

Abdellah RAHMANI

PhD candidate at EPFL, Switzerland

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About me

I am a PhD candidate at EPFL, advised by Prof. [Pascal Frossard](#). My research studies how sequence models should structure and allocate their computation. My early work includes Meta-GNN, an efficient meta-learned model that personalizes to each epileptic patient from only a few adaptation steps. Asking where such efficiency ultimately comes from led me to structure: I developed the CASTOR, FANTOM, and Cheesefill frameworks to learn the causal structure underlying multivariate time series, in the challenging regime of non-stationary and missing data. Currently, my research brings conditional structure to foundation models: I build deep recurrent architectures that leverage dynamic routing and test-time scaling, so that computation grows with the difficulty of the reasoning problem. I am also an incoming Student Researcher at [Google DeepMind](#) London, focusing on in-context and prequential learning for LLMs. Before joining EPFL, I completed my Master's degree at ENS Paris-Saclay ([Mathematics](#), [Vision](#), [Learning: MVA](#)) and gained industry experience as a research intern at [Deezer Research](#).

Education

2022–ongoing **EPFL**, Lausanne

PhD, Electrical Engineering: Artificial intelligence and Machine learning.

2020–2021 **Ecole Normale Supérieure Paris-Saclay**, Paris, [Highest honors](#)

Master of Science : Mathématiques, Vision et Apprentissage (MVA - Mathematics, Vision and Learning).

2019–2020 **Ecole Centrale Lille**, Lille, [Part of top 10% students among 280 students](#)

Final year of engineering : Artificial intelligence and Data science, ex DAD (Research oriented).

In addition to advanced courses of Mathematics at the university of Lille in parallel of my engineering curriculum: Functional analysis, Stochastic processes, High dimensional probability.

2017–2019 **Ecole Centrale Casablanca**, Casablanca

Master of Engineering,

Major: Data Science and AI (Research oriented).

2014–2017 **Preparatory classes**, Meknes, [Rank: 104/3000](#)

Intensive program to enter top engineering schools.

Professional Experience

08/2026-12/2026 **Student researcher**, *Google DeepMind*, London

In-context and prequential learning for LLMs.

08/2022-Ongoing **Doctoral assistant**, *EPFL*, Lausanne

Efficient AI Reasoning Across Modalities: Causal Structure Learning for Temporal Data and Dynamic Routing for Text and Vision.

04/2021-09/2021 **Research scientist Intern**, *Deezer Research*, Paris

Enhancing podcast recommendations through users' music consumption patterns.

The research project consisted on implementing new approaches for cross domain recommendation such as Domain Adaptation and Meta Learning.

04/2020-09/2020 **Applied Scientist Intern**, *Silex France*, Antibes

Exploring Graph Embedding for Search Engines.

10/2019-03/2020 **Part-time Researcher**, *Cristal*, Lille

Applied domain adaptation techniques to deep learning models for the task of image manipulation detection (under the guidance of [Patrick Bas](#)).

04/2019-08/2019 **Data Science Intern**, *Forafic Maroc*, Casablanca

Crop Identification from Satellite Images Using Machine Learning Models.

Publications

[1] [A. Rahmani*](#), [S. McLeish*](#), [A. Li](#), [S. Lotfi](#), [P. Frossard](#) and [M. Goldblum](#), Library of Layers (LoL): Adaptively Ordering Transformer Layers at Runtime. *Submitted 2026*

[2] [A. Rahmani](#), and [P. Frossard](#), Causal Heteroscedastic Structure Learning From Incomplete Temporal data. *NeurIPS 2026*

- [3] **A. Rahmani**, and *P. Frossard*, Flow based approach for Dynamic Temporal Causal models with non-Gaussian or Heteroscedastic Noises. *NeurIPS 2025*
- [4] **A. Rahmani**, and *P. Frossard*, Causal Temporal Regime Structure Learning. *AISTATS 2025*
- [5] **A. Rahmani**, *A. Venkitaraman*, and *P. Frossard*, A Meta-GNN approach to personalized seizure detection and classification. *ICASSP 2023*

Kaggle Challenges

- 03/2021-04/2021 **Kaggle Data Challenge 2**, *ENS Paris Saclay*, Paris
Classification of DNA Sequence Regions for Transcription Factor Binding Sites Using Kernel Methods, **Rank: 3/70** (70 groups of 3)
- 03/2020-04/2020 **Kaggle Data Challenge 1**, *Centrale Lille*, Lille
Hand Gesture Recognition Using Deep Learning Techniques (CNN, RNN, etc.), **Rank: 1/10** (10 groups of 4)

Awards

1. **EPFL**, *EPFL Doctoral School*, *Lausanne*, Doc.Mobility grant (7,000 CHF) for Deep recursive models in collaboration with Columbia University.
2. **EPFL**, *EDEE Doctoral program*, *Lausanne*, 3rd place in my thesis in 180 seconds.
3. **EPFL**, *LTS4*, *Lausanne*, Master's valorization scholarship (10,000 CHF).
4. **Ecole Centrale de Lille**, *School of Engineering*, *Lille*, Top 10% among 280 students at Ecole Centrale Lille.
5. **Fleurs d'orangers**, *High school*, *Meknes*, Best student at high school among 30 students.

Languages

Arabic: Mother tongue
English: Fluent
French: Fluent

Community service

Reviewer in: NeurIPS, ICLR and ICASSP
Evaluator in: ELLIS PhD program

Teaching Experience

1. Network Machine learning, EPFL, Spring 2023, Spring 2024 and Spring 2025.
2. Optimization, EPFL, Fall 2023.
3. Linear algebra, EPFL, Fall 2022.
4. Supervised 2 student projects on representation learning from EEG data, EPFL, Spring 2023 and Fall 2024.
5. Signal processing, EPFL, Fall 2025.

Software

Programming Languages: Python, MySQL, Java, Spark
Deep learning Frameworks: PyTorch, JAX, Keras
Misc: Unix, Git, Docker

Hobbies

Sports: Football, Padel, Swimming
Arts: Part time Rapper, Writing